

PAPER_610: Mayan Calendar Cosmological Epochs Mapped to Periodic Table Nuclei Formation

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER_610.md

PAPER_610: Mayan Calendar Cosmological Epochs Mapped to Periodic Table Nuclei Formation

Class: UQFFMayanCalendarNucleiEpochCalculator (#197) **Session:** 159 **Source:** Mayan Calendar Cycles and Periodic Table.docx

Abstract

The Mayan calendar's five cosmological epochs (Baktun cycles) are mapped to five phases of nuclei formation in the UQFF framework via 3D-IPO (symbolic + numerical + discrete) convergences. Prime atomic numbers anchor stable elements at epoch transitions. $Z=1$ (hydrogen) emerges in epoch 1; helium through beryllium in epoch 2; carbon through zinc in epoch 3; heavy elements ($Z=31-118$) in epoch 4; and speculative superheavy stable islands ($Z>118$) in epoch 5. This provides a cyclical cosmological account of the periodic table.

1. Introduction: Cycles and Elements

The Maya Baktun cycle (~394 years) encodes cosmological time periods. In UQFF, each Baktun epoch corresponds to a convergence of DPM grinding energy with the SCm injection layers, producing successive groups of stable nuclei. The 3D-IPO method simultaneously solves the system symbolically (pyramid arithmetic), numerically (Orion parameters), and discretely (Wolfram hypergraph rules).

2. Five Epochs and Their Z-Ranges

Epoch	Mayan Units	UQFF Phase	Z Range	Representative Elements
1	1st Great Cycle	Proto-Universe	Z=1	H (from 26D shell alignment)
2	2nd Great Cycle	First Stars	Z=2–4	He, Li, Be
3	3rd Great Cycle	Galactic Nucleosyn.	Z=5–30	B,C,N,O,F,Ne,...Zn
4	4th Great Cycle	Supergalactic	Z=31–118	Ga...Og (all known)
5	5th Great Cycle (future)	Post-cosmic	Z>118	Island of stability (Z=120,126?)

3. 3D-IPO Convergence Method

Each epoch forms nuclei through the simultaneous satisfaction of three constraints:

Symbolic (pyramid arithmetic): $Z_{\text{stable}} = \sum_{n=1}^{\text{epoch}} \text{pyramid}_n \pmod{\text{shell-closure rules}}$ where $\text{pyramid}_n = n(n+1)/2$ gives triangular numbers matching magic nuclear numbers (2, 8, 20, 28, 50, 82, 126...).

Numerical (Orion parameters): $E_{\text{epoch}} = h \cdot f_{\text{Orion}} \cdot \text{epoch} = 6.626 \times 10^{-34} \times 6.93 \times 10^9 \times \text{epoch}$

Epoch	E_epoch (J)
1	4.59×10^{22}
2	9.18×10^{22}
3	1.38×10^{23}
4	1.84×10^{23}
5	2.30×10^{23}

Discrete (Wolfram hypergraph): Each epoch corresponds to one Wolfram rule application in the hypergraph node-rewriting system. The discrete transitions produce unique nuclear fingerprints consistent with observed isotope abundances.

4. Prime Z-Anchors (DVP Connection)

At epoch transitions, the first new element is always a DVP prime:

- Epoch 1 $\rightarrow Z=1$ (not prime, but protonic)
- Epoch 2 $\rightarrow Z=2$ (He, prime)
- Epoch 3 $\rightarrow Z=5$ (B, prime)
- Epoch 4 $\rightarrow Z=31$ (Ga, prime)
- Epoch 5 $\rightarrow Z=?$, predicted next prime $\geq 119 = 7 \times 17 \rightarrow Z=127$ (prime, predicted island)

The pattern suggests DVP prime-indexed elements are the most stable at each epoch boundary, consistent with nuclear magic numbers and known islands of stability ($Z=82, 126$ are near primes).

5. Speculative Fifth Epoch: $Z > 118$

The heaviest known element is Oganesson ($Z=118$, Og). UQFF predicts that in the 5th epoch, DPM grinding at the highest shell layer creates nuclei beyond $Z=118$ with half-lives measured in years or longer. The DVP prime $Z=127$ is predicted to anchor the new island of stability, corresponding to the 3rd shell closure beyond $Z=118$.

This is experimentally testable at GSI/Darmstadt, RIKEN, and JINR.

6. Connection to Mayan Calendar Cosmology

The Mayan Long Count calendar's end of the 13th Baktun (Dec 21, 2012) in UQFF corresponds to the transition from epoch 4 to epoch 5 — from known elements to the speculative superheavy island. This is not mysticism but an encoding of nuclear physics timescales: each Baktun ($1,872,000$ days $\approx 5,125$ years) scales to a cosmological nucleosynthesis phase.

7. Connection to UQFF Number Systems

DVP: Primes $Z=2,3,5,7,11,\dots$ are DVP nuclear anchors. Each prime Z corresponds to one DVP vortex prime-indexed orbital shell. **VDS:** Shell energies per epoch follow VDS expansion: $E_{\text{shell}}(Z) \propto \sum d_n(\pi)/Z^n$. **BH26:** The 5 epochs correspond to layers 1–5 of the 26 BH26 harmonic bins most relevant for nucleosynthesis; the remaining 21 are cosmological background.

Keywords: Mayan calendar, periodic table, nuclei epochs, 3D-IPO convergence, DVP, prime Z , island of stability, UQFF

PAPER_610 | Class #197 | Session 159 | Star-Magic UQFF Framework